Home walkthrough: kitchen4_lighter

10/25/2026, 1:29:43 PM · This patient (walker) · knees 50.0 cm, seated reach 60.0 cm

For the occupational therapist to review: a flag list, not a diagnosis. Measurements come from a 3D reconstruction of the home (Gaussian splat) scaled to metres; anything marked imagined is generated and never used for a number; **OT to verify** marks what the scan can't settle.

Expected toe catches per month

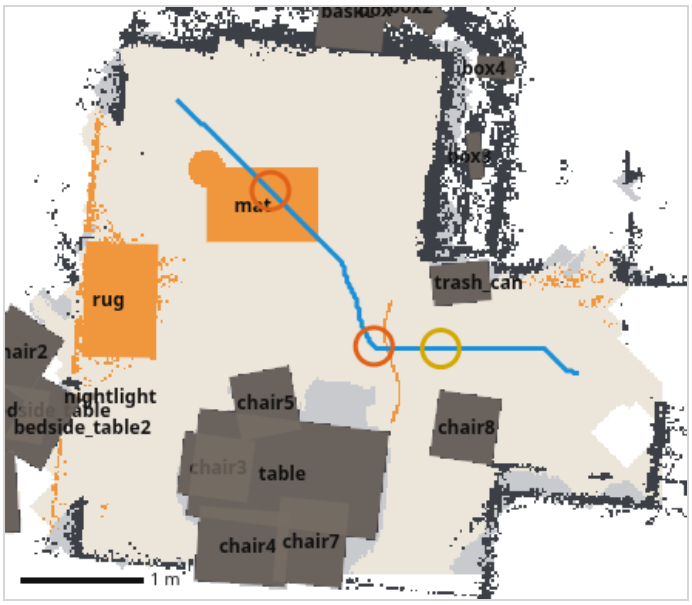
1.3

likely 1.2–1.3 · original layout 1.1 (**+0.2**)

200 simulated trips per route, gait "older_adult_walker". Toe-contact rate calibrated to published data (MTC 14.1 ± 8.3 mm lognormal shape; ~1 in 578 strides contact a 1 cm obstacle (Frontiers 2014)); darkness multiplier 1.5 (assumption). Range = measured height error + simulation noise.

Route	Trips/night	Contacts/month	Length
main route (night)	2	1.3 (1.2–1.3)	4.4 m

Layout and route





Map: grey = walls/furniture, orange = floor bumps (rugs, thresholds, cables), blue line = route, rings = hazards, blue furniture = moved, dashed = where it stood, light grey floor = scanned too thinly to trust (not walked).

Measured hazards on the route (main route)

What	Measurement	Along route	Contacts/month	
tight passage	3.5 cm each side of the walker	0.70 m	–	**OT to verify**
folded rug corner	4.2 cm high	1.79 m	1.2	**OT to verify**
cable	0.5 cm high **(estimated: classified, not measured)**	0.98 m	0.1	**OT to verify**

Transfers, reach and emergencies

Verdict	Check	Basis
fail	chair height 34 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair2 height 33 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair3 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair4 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair5 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair7 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair8 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
OT to verify	on the floor at the tight passage: no phone marked	farther than 3 m to crawl; consider a wearable alert
OT to verify	on the floor at the folded rug corner: no phone marked	farther than 3 m to crawl; consider a wearable alert
OT to verify	on the floor at the cable: no phone marked	farther than 3 m to crawl; consider a wearable alert

Changes in this layout

#	Change	Cost	Time	Product
none: the home as filmed				

Recommended fixes, ranked

#	Fix	Toe catches/month saved	Passage gained	Cost	Time	Effort
1	**Remove the kitchen mat**	**−1.08** (95%)	–	free	2 min	easy
2	**Tape the mat's edges down**	**−1.07** (94%)	–	$10	10 min	easy

Each fix is simulated on its own from the layout above (200 trips per route); savings are not additive. Ranked by kitchen planner (B200 layout search + critic): each change alone on this layout, by toe catches/month saved (the same calculation as above), ties broken by the search objective (toe catches + simulated snags + brushes).

Planner options

Option	Toe catches/month	Cost · time	Effort	Changes from the scanned room
Easiest: most toe catches removed per unit of effort	0.07	free · 2 min	1.0	Remove the kitchen mat
Keep my mat: most toe catches removed per unit of effort without removing it	0.07	$10 · 10 min	1.0	Tape the mat's edges down
Safest: fewest toe catches found	0.00	$8 · 58 min	5.0	Move chair8 to (0.40, -0.20), turned 180 deg; Remove the kitchen mat; Tape the cord flat

Each option re-run on fresh simulated trips (200 per route), not the search's own draws.

Furniture the walker could push (simulated)

Object	Event	Route	Approach	Contact	P per touch/crossing (95% CI)	Per month
chair8	moves ≥ 2 cm	main route	9° · 0.50 m/s	28% of trips touch it	91% (83%–95%, n=95)	15.3
chair8	tips > 30°	main route	9° · 0.50 m/s	28% of trips touch it	0.0% (0.0%–3.9%, n=95)	0.0
mat	slides ≥ 2 cm	main route	45° · 0.50 m/s	every trip crosses it	66% (58%–74%, n=127)	39.7
mat	corner folds over	main route	45° · 0.50 m/s	every trip crosses it	2.4% (0.8%–6.7%, n=127)	1.4

SIMULATED (MuJoCo batch, physics/batch_mujoco.py): shapes from the scan; masses, frictions and corner stiffness sampled from assumed ranges, not measured; not yet checked against the real walker. Approach angle 0° = straight at the edge.

Generated visualisations of this layout

- imagined, not measured

Cosmos Transfer videos: imagined, for the family; hazard markers on them are the measured ones.

Methods and limits

- Measurements close to a limit are marked "borderline, OT to verify".
- Thin floor items (cables) are classified, not measured: their height is a typical value.
- Floor and walls under or behind furniture that was moved were never filmed; they are shown imagined (tinted) and not measured.
- Toe contacts are not falls; the numbers compare layouts for this person and route, they are not a fall probability.